THE EREBUS
SUB-GLACIAL RESEARCH

SITUATION REPORT

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Falling Out of Reality: A Visual Guide to the Mechanics and Geometry of Anomalous Sub-Spaces

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Executive Summary: The phenomenon of "noclipping"—falling through the dimensional boundary of baseline reality into an endless, empty maze—has been confirmed by deep-ice seismology at Outpost 31. To brief incoming containment staff, this document connects the dots between three confirmed mechanisms of the anomaly: (1) Spacetime elasticity (how reality tears under localized stress), (2) Quantum Superposition (how the sub-dimension generates architecture upon observation), and (3) Hyperbolic Geometry (the mathematical reason why trapped subjects cannot navigate out). Using accessible language and visual models, we build a complete theoretical baseline for how this extradimensional trap functions, and why rescue operations remain suspended indefinitely.

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1. Introduction: Connecting the Dots

How does a subject simply fall out of the world? And once they do, why does the telemetry indicate the space they enter goes on forever, changing every time they turn around?

To understand the sub-dimension (Internal Designation: "The Backrooms") from a containment perspective, personnel must understand three distinct phases of the anomaly using our current physical models:

1. **The Entry:** Breaking the fabric of reality.
2. **The Environment:** A world that only exists when observed.
3. **The Trap:** The geometry that makes escape mathematically impossible.

This briefing breaks down these three phases, utilizing visual models to make the underlying mathematics intuitive for non-specialist field agents.

2. Phase 1: The Entry (Noclipping)

In early computational simulations, collision errors allow a simulated object to pass through a solid boundary. In the physical world, we model this by observing the fabric of spacetime.

2.1. The Rubber Sheet Analogy

Spacetime behaves structurally like a tightly stretched rubber sheet. Massive objects weigh it down, creating the curves we experience as gravity. But what happens if an anomalous kinetic force pushes on this rubber sheet too hard in a concentrated area?

Every physical material has a "Yield Stress"—a maximum limit of pressure it can withstand before tearing. If an anomalous force impacts our spacetime fabric, and the pressure (σ) exceeds the fabric's ultimate tensile strength (σ_{crit}), the "rubber sheet" of reality tears.

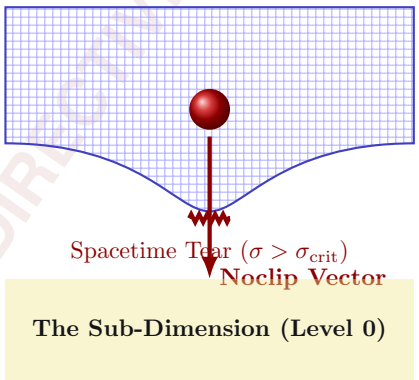

Figure 1: The Mechanics of Entry. A localized anomaly pushes against the "rubber sheet" of baseline reality. When the pressure exceeds the fabric's limits, a temporary topological tear opens, dropping the subject into the sub-dimension below.

As shown in Fig. 1, the subject falls through this temporary microscopic shear zone. The restorative pressure of the cosmological constant causes reality to quickly snap back together like a healing wound, trapping the subject on the other side.

3. Phase 2: The Self-Building Environment

Once inside, audio logs recovered via tethered drones consistently report an endless series of empty, mono-yellow

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office rooms. But how can an infinite space exist underneath our universe without collapsing under its own gravity? The answer lies in quantum mechanics and algorithmic procedural generation.

3.1. Schrödinger's Maze

In theoretical computer modeling, to conserve memory, a rendering engine only calculates the geometry that the observer is currently looking at. The sub-dimension operates on a similar thermodynamic principle, relying on quantum superposition to conserve universal energy.

Before a trapped subject looks at a room, the room exists only as a blurry cloud of mathematical probabilities. It could be an empty hallway, it could be a dead end, or it could be a pillar.

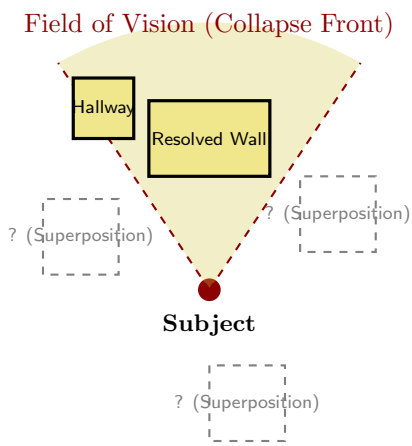

Figure 2: Procedural Generation of Reality. The space is not pre-built. As the subject looks around, their "Field of Vision" forces the quantum blur to collapse into solid walls and carpet. When they turn around, the space behind them vanishes back into probability.

The moment the subject's visual cortex processes the space, the "wavefunction collapses," and the universe instantly builds physical matter (Fig. 2). This is why the rooms feel completely random and disjointed—because they are literally generating on the fly, second by second, driven by a stochastic Wang-tiling algorithm.

4. Phase 3: The Geometry of Getting Lost

If a field agent runs in a perfectly straight line for days, why do they never hit an outer boundary or find the exact spot they fell in? This is governed by **Hyperbolic Geometry**.

4.1. The Expanding Circle

In our baseline Euclidean world, if you draw a circle on the ground, the area inside that circle grows at a predictable, steady rate as you make the circle wider.

However, imagine drawing a circle on a saddle-shaped curve. Because the surface curves away from itself, there is exponentially more surface area on the edges than in the middle. The sub-dimension is a hyperbolic manifold (curved outward). As an agent walks away from their drop-point, the amount of space around them multiplies exponentially.

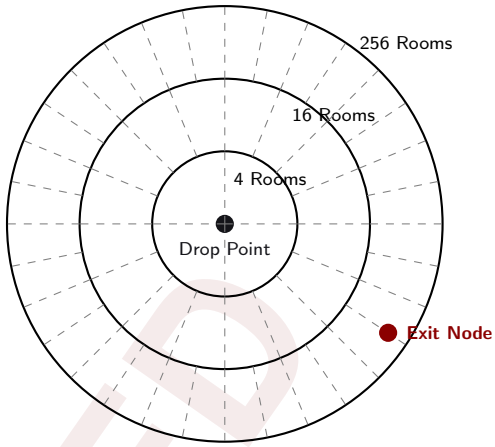

Figure 3: Hyperbolic Expansion. A top-down topological projection. Moving just one step outward doesn't simply add a few rooms; it squares the available volume. Finding a single "Exit Node" becomes a statistical impossibility as the search radius grows.

4.2. The Math of Confinement

We graph this relationship to demonstrate why Erebus Directive Command has banned all rescue operations.

In baseline reality, searchable area grows normally ($Volume = \text{Radius}^3$). But because the sub-dimension generates hyperbolically based on the observer's movement, the number of new rooms being generated outpaces physical movement ($Volume = e^{\text{Radius}}$).

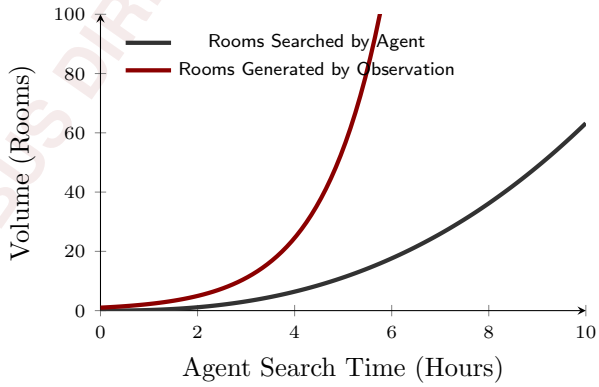

Figure 4: The Confinement Trap. The anomaly generates new topology (red line) vastly faster than human physiology permits traversal (grey line). The agent can never catch up to the expanding perimeter.

As seen in Fig. 4, human biology cannot search the space before it multiplies. The anomaly mechanically outpaces the observer. Therefore, the probability of a subject randomly intersecting with a reverse-noclip exit node approaches 0% within 72 hours of entry.

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5. Conclusion

The anomaly designated as the "Backrooms" is not a supernatural space, but a rigorously defined topological cage. By bridging spacetime mechanics with quantum rendering and hyperbolic geometry, we confirm that the sub-dimension is mathematically inescapable. All personnel at Outpost 31 are reminded to maintain a minimum 50-meter distance from confirmed weak-points in the containment grid.

Acknowledgements

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